

```
In [1]: graph = {
    "A": ["B", "C"],
    "B": ["D", "E"],
    "C": ["F"],
    "D": [],
    "E": ["G"],
    "F": [],
    "G": []
}

def dls(node, goal, limit, path):

    if node == goal:
        return "FOUND", path

    if limit == 0:
        return "CUTOFF", None

    cutoff = False

    for child in graph[node]:

        if child in path:
            continue

        status, result = dls(child, goal, limit - 1, path + [child])

        if status == "FOUND":
            return "FOUND", result

        if status == "CUTOFF":
            cutoff = True

    if cutoff:
        return "CUTOFF", None

    return "FAILURE", None

def iddfs(start, goal, max_depth):

    limits_attempted = []

    for limit in range(max_depth + 1):

        limits_attempted.append(limit)

        status, path = dls(start, goal, limit, [start])

        if status == "FOUND":
            return status, path, limits_attempted

    return "CUTOFF", None, limits_attempted
```

```
print("DLS RESULTS")

for limit in [1, 2, 3, 4]:

    status, path = dls("A", "G", limit, ["A"])

    print("Limit:", limit)
    print("Status:", status)
    print("Path:", path)
    print()

print("IDDFS RESULTS")

for max_depth in [1, 2, 3, 4]:

    status, path, limits = iddfs("A", "G", max_depth)

    print("Maximum Depth:", max_depth)
    print("Status:", status)
    print("Path:", path)
    print("Limits Attempted:", limits)
    print()
```

DLS RESULTS

Limit: 1

Status: CUTOFF

Path: None

Limit: 2

Status: CUTOFF

Path: None

Limit: 3

Status: FOUND

Path: ['A', 'B', 'E', 'G']

Limit: 4

Status: FOUND

Path: ['A', 'B', 'E', 'G']

IDDFS RESULTS

Maximum Depth: 1

Status: CUTOFF

Path: None

Limits Attempted: [0, 1]

Maximum Depth: 2

Status: CUTOFF

Path: None

Limits Attempted: [0, 1, 2]

Maximum Depth: 3

Status: FOUND

Path: ['A', 'B', 'E', 'G']

Limits Attempted: [0, 1, 2, 3]

Maximum Depth: 4

Status: FOUND

Path: ['A', 'B', 'E', 'G']

Limits Attempted: [0, 1, 2, 3]

In []: